# 1 Formula for the inverse of a matrix

**Exercise 1.1.** Let $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$. Prove that

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} A_{22} & A_{12} \\ -A_{21} & A_{11} \end{bmatrix} \tag{1.2}$$

Using the definition of the $(i,j)$-cofactor $C_{ij} = (-1)^{i+j} \det A[i|j]$ and arranging these as a $2 \times 2$ matrix, we find

$$\begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} A_{22} & -A_{21} \\ -A_{12} & A_{11} \end{bmatrix}.$$

Therefore, we see that we can identify the matrix in Equation (1.2) as the transpose of the matrix of cofactors of $A$.

**Definition 1.3** (Adjugate matrix)**.** Let $A$ be an $n \times n$ matrix. The transpose of the matrix of cofactors of a $A$ is called the *adjugate* of $A$, denoted adj $A$. [1]

Equation (1.2) can therefore be written as

$$A^{-1} = \frac{1}{\det A} \text{adj } A.$$

We will prove below that this formula holds for any $n \times n$ invertible matrix. This formula is useless for practical calculations (calculating $A^{-1}$ by row-reduction is far more efficient), but sometimes it is useful to have an explicit formula for $A^{-1}$ for theoretical purposes.

**Proposition 1.4.** *Let $A$ be an invertible $n \times n$ matrix. Then*

$$A^{-1} = \frac{1}{\det A} adj\, A.$$

*Proof.* We need to show that $\frac{\text{adj } A}{\det A} A = I$. Multiplying through by $\det A$, this becomes $A(\text{adj } A) = (\det A) I$, which is what we will now prove. First, recall the cofactor expansion for $\det A$ along the $j$th column:

$$\det A = \sum_{i=1}^{n} A_{ij} C_{ij} = \sum_{i=1}^{n} A_{ij} (-1)^{i+j} \det A[i|j].$$

We now claim that if $j \neq k$, then the expression

$$\sum_{i=1}^{n} A_{ik} C_{ij} = 0.$$

[1] The matrix adj $A$ is also sometimes called the *classical adjoint* of $A$. We will see another adjoint when we discuss inner product spaces, which has nothing to do with adj $A$. We reserve the term "adjoint" for the latter.

To see this, let $B$ be the matrix obtained from $A$ by replacing the $j$th column of $A$ by the $k$th column of $A$. Since $B$ has two equal columns, $\det B = 0$. Since $B[i|j] = A[i|j]$, computing $\det B$ by cofactor expansion along the $j$th column gives

$$\begin{aligned} 0 = \det B &= \sum_{i=1}^{n} B_{ij}(-1)^{i+j} \det B[i|j] \\ &= \sum_{i=1}^{n} B_{ik}(-1)^{i+j} \det A[i|j] \\ &= \sum_{i=1}^{n} B_{ik} C_{ij}, \end{aligned}$$

as claimed. These two properties of cofactors can be summarized as

$$\sum_{i=1}^{n} A_{ik} C_{ij} = \delta_{kj}(\det A). \tag{1.5}$$

Since left hand side of this equation is equal to $\sum_{i=1}^{n} A_{ik} C_{ji}^T = \sum_{i=1}^{n} C_{ji}^T A_{ik}$, we see that Equation (1.5) is the $kj$-component of the matrix equation

$$(\operatorname{adj} A)A = (\det A)I. \tag{1.6}$$

□

**Example 1.7.** Consider the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$. We first calculate all 9 cofactors of $A$:

$$\begin{aligned} C_{11} &= \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, \quad C_{12} = -\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, \quad C_{13} = \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5 \\ C_{21} &= -\begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, \quad C_{22} = \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad C_{23} = -\begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7 \\ C_{31} &= \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, \quad C_{32} = -\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{33} = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3. \end{aligned}$$

The adjugate matrix is therefore $\operatorname{adj} A = \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{bmatrix}$. Since we have already calculated the cofactors, it is now straightforward to calculate $\det A$:

$$\det A = A_{11}C_{11} + A_{12}C_{12} + A_{13}C_{13} = 2(-2) + 1(3) + 3(5) = -4 + 18 = 14$$

Therefore

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -1/2 & -3/14 \end{bmatrix}.$$

**Exercise 1.8.** Use the formula $A^{-1} = (\det A)^{-1}\text{adj } A$ to compute the inverse of the matrix

$$A = \begin{bmatrix} 5 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & -1 \end{bmatrix}$$

# 2 Cramer's rule

**Theorem 2.1** (Cramer's Rule)**.** *Let $A$ be an invertible $n \times n$ matrix. For any $\mathbf{b} \in \mathbb{R}^n$, the unique solution $\mathbf{x}$ of the linear system $A\mathbf{x} = \mathbf{b}$ has entries given by*

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \ldots, n, \tag{2.2}$$

*where $A_i(\mathbf{b})$ is the matrix obtained from $A$ by replacing the $i$th column of $A$ by $\mathbf{b}$. The formula in Equation* (2.2) *for computing the entries of $\mathbf{x}$ is called* Cramer's rule.

**Exercise 2.3.** If $A$ is invertible then the linear system $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. Use the formula for $A^{-1}$ from Proposition 1.4 to prove Theorem 2.1.

*Remark* 2.4. Cramer's rule is very inefficient for solving linear systems, as computing just *one* determinant takes about as much work as solving $A\mathbf{x} = \mathbf{b}$ by row-reduction. Cramer's rule is mostly useful as a theoretical tool.

**Example 2.5.** Consider the system

$$\begin{aligned} 3x_1 - 2x_2 &= 6 \\ -5x_1 + 4x_2 &= 8. \end{aligned}$$

The coefficient matrix is $A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}$. Replacing each column of $A$ with $\mathbf{b} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$ gives $A_1(\mathbf{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}$ and $A_2(\mathbf{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$. Calculating determinants

$$\det A = 2, \quad \det A_1(\mathbf{b}) = 40, \quad \det A_2(\mathbf{b}) = 54,$$

Cramer's rule gives the unique solution

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 40/2 \\ 54/2 \end{bmatrix} = \begin{bmatrix} 20 \\ 27 \end{bmatrix}.$$

**Exercise 2.6.** Use Cramer's rule to solve the system

$$\begin{aligned} -5x_1 + 2x_2 &= 9, \\ 3x_1 - x_2 &= -4. \end{aligned}$$

**Exercise 2.7.** Use Cramer's rule to solve the system

$$\begin{aligned} x_1 + x_2 &= 3, \\ -3x_1 + 2x_3 &= 0, \\ x_2 - 3x_3 &= 2. \end{aligned}$$